

Instructions: Use graph paper or the grids provided by your teacher. Label points with capital letters, write ordered pairs in parentheses, and show your reasoning.

Practice 1: The coordinate plane

Name: _____ Date: _____

1. In an ordered pair, what does the first number tell you? What does the second number tell you? Explain how you would plot $(4, 2)$.

Work space

2. Draw a coordinate plane with axes from 0 to 6. Plot $A(2, 5)$ and $B(5, 1)$ and label them.

Work space

Practice 2: Reading coordinates

Name: _____ Date: _____

1. Give the ordered pair for a point 3 units right and 6 units up from the origin.

Work space

2. Point C is at $(7, 4)$. State its x-coordinate and y-coordinate, then describe how to reach it from $(0, 0)$.

Work space

Practice 3: Plotting points

Name: _____ Date: _____

1. Plot $D(1, 1)$, $E(1, 5)$, $F(5, 5)$, and $G(5, 1)$ on a coordinate grid. Connect them in order. What shape is formed?

Work space

2. Plot $H(2, 6)$, $I(4, 6)$, and $J(4, 3)$. Which point is highest? Which two have the same x-coordinate?

Work space

Practice 4: Horizontal and vertical movement

Name: _____ Date: _____

1. Start at $K(2, 3)$. Move 4 units right. Write the new ordered pair and explain which coordinate changed.

Work space

2. Start at $L(6, 5)$. Move 3 units down. Write the new ordered pair and explain why.

Work space

Practice 5: Translating points

Name: _____ Date: _____

1. Translate $M(1, 4)$ 3 units right and 2 units up. Give M' and show the coordinate changes.

Work space

2. A point $N(7, 2)$ is translated 5 units left and 1 unit down. Give N' .

Work space

Practice 6: Translating figures

Name: _____ Date: _____

1. Triangle PQR has vertices $P(1, 1)$, $Q(3, 1)$, and $R(2, 4)$. Translate it 2 units right. List P' , Q' , and R' .

Work space

2. Rectangle $STUV$ has vertices $(2, 2)$, $(6, 2)$, $(6, 4)$, $(2, 4)$. Translate it 1 unit left and 3 units up. List all four new vertices.

Work space

Practice 7: Reflections across the x-axis

Name: _____ Date: _____

1. Reflect $A(3, 5)$ across the x-axis. Give A' and describe the rule.

Work space

2. Reflect triangle vertices $B(1, 2)$, $C(4, 2)$, and $D(2, 6)$ across the x-axis. List the image coordinates.

Work space

Practice 8: Reflections across the y-axis

Name: _____ Date: _____

1. Reflect $E(6, 2)$ across the y-axis. Give E' and state which coordinate changes.

Work space

2. Reflect rectangle vertices $(1, 1)$, $(4, 1)$, $(4, 3)$, $(1, 3)$ across the y-axis. List the four image vertices.

Work space

Practice 9: Reflection reasoning

Name: _____ Date: _____

1. Point F is $(0, 4)$. What happens when it is reflected across the y -axis? Explain.

Work space

2. A point and its reflection across the x -axis are $(5, 3)$ and $(5, \square)$. Fill the blank and explain.

Work space

Practice 10: Quarter-turn rotations

Name: _____ Date: _____

1. A toy arrow points up. After a quarter-turn clockwise, which direction does it point? After another quarter-turn clockwise?

Work space

2. A square corner marker has vertices $(1, 1)$, $(3, 1)$, $(3, 3)$, $(1, 3)$. Sketch a quarter-turn clockwise around its center $(2, 2)$ and label the image.

Work space

Practice 11: Rotation around a point

Name: _____ Date: _____

1. Point G is 2 units right of center $O(4, 4)$. After a quarter-turn clockwise around O , where is G' ?

Work space

2. Point H is 3 units above center $O(5, 5)$. After a quarter-turn counterclockwise around O , where is H' ?

Work space

Practice 12: Distance on a grid

Name: _____ Date: _____

1. Find the horizontal distance between $J(2, 4)$ and $K(7, 4)$.

Work space

2. Find the vertical distance between $L(3, 1)$ and $M(3, 6)$.

Work space

Practice 13: Perimeter from coordinates

Name: _____ Date: _____

1. A rectangle has vertices $(1, 1)$, $(6, 1)$, $(6, 4)$, $(1, 4)$. Find its side lengths and perimeter.

Work space

2. A square has vertices $(2, 2)$, $(5, 2)$, $(5, 5)$, $(2, 5)$. Find its perimeter. Explain how the coordinates help.

Work space

Practice 14: Area from coordinates

Name: _____ Date: _____

1. Find the area of the rectangle with vertices $(2, 1)$, $(8, 1)$, $(8, 4)$, $(2, 4)$.

Work space

2. A square runs from $x = 3$ to $x = 7$ and $y = 2$ to $y = 6$. Find its side length and area.

Work space

Practice 15: Congruent figures

Name: _____ Date: _____

1. Two rectangles have side lengths 4 units and 7 units. Are they congruent? What does congruent mean?

Work space

2. A triangle is translated 5 units right. Does its size or shape change? Explain why the original and image are congruent.

Work space

Practice 16: Comparing transformations

Name: _____ Date: _____

1. Which transformation slides a figure without turning or flipping it? Describe it.

Work space

2. Which transformation changes $(4, 2)$ to $(-4, 2)$? Name the line of reflection and explain.

Work space

Practice 17: Multi-step transformations

Name: _____ Date: _____

1. Start with $A(2, 3)$. Translate 2 units right, then reflect across the y-axis. Give the final coordinate.

Work space

2. Start with $B(5, 1)$. Reflect across the x-axis, then translate 1 unit left and 2 units up. Give the final coordinate.

Work space

Practice 18: Coordinate word problems

Name: _____ Date: _____

1. On a map, the library is $(2, 5)$ and the park is $(2, 1)$. How far apart are they if each grid unit is 1 block?

Work space

2. A robot at $(3, 4)$ moves 4 units east and 3 units south. Where does it stop?

Work space

Practice 19: Design on a coordinate grid

Name: _____ Date: _____

1. Create a rectangle with width 3 and height 2 whose lower-left vertex is $(1, 1)$. List all vertices.

Work space

2. Create a triangle with vertices $(1, 2)$, $(4, 2)$, $(1, 5)$. Find its horizontal and vertical side lengths.

Work space

Practice 20: Mixed review

Name: _____ Date: _____

1. Rectangle $WXYZ$ has vertices $(1, 2)$, $(5, 2)$, $(5, 5)$, $(1, 5)$. Translate it 2 units left, then reflect it across the x-axis. List the final vertices.

Work space

2. Explain one similarity and one difference between a translation and a reflection. Give a coordinate example for each.

Work space